5.5 Performing Function Operations

Name _____ Period ____ Date _____

Algebra II 5.5 · Student sheet

I can…

- add and subtract functions.
- multiply and divide functions.
- find the domain of a combined function and evaluate it.

Vocabulary · fill in the missing word

(f + g)(x) — _____

(f − g)(x) — _____

(fg)(x) — _____

(f/g)(x) — _____

Domain of a combined function — the _____ in the domain of BOTH f and g (and $g(x) \neq 0$ for $\frac{f}{g}$)

Part 1 · Profit table

A T-shirt business has revenue $R(x) = 15x$ and cost $C(x) = 4x + 200$, where x is the number of shirts sold.

x	R(x) = 15x	C(x) = 4x + 200	R(x) − C(x)
0			
20			
40			
60			

Write one function for the profit: P(x) = R(x) − C(x) = _____

Part 2 · Notes and examples (fill in each note, then work the example)

Sum and difference: (f + g)(x) = _____ (f − g)(x) = _____

Domain of a combined function: The domain of a combined function is the x-values in the domain of _____ f and g.

Example 1

Let $f(x) = 3\sqrt{x}$ and $g(x) = -10\sqrt{x}$. Find (f + g)(x) and (f − g)(x) and state the domain of each.

Example 2

Let $f(x) = 3x^3 - 2x^2 + 5$ and $g(x) = x^3 - 3x^2 + 4x - 2$. Find (f − g)(x).

Product and quotient: (fg)(x) = _____ (f/g)(x) = _____, where _____

Domain of f/g: For f/g, also leave out every x that makes _____.

Example 3

Let f(x) = 6x and g(x) = x^($\frac{3}{4}$). Find (fg)(x) and ($\frac{f}{g}$)(x) and state the domain of each.

Example 4

Let f(x) = $x^2 - 1$ and g(x) = $\sqrt{x}$. Find (f + g)(4) and (fg)(9).

Example 5

A dog groomer's revenue is R(x) = 45x and cost is C(x) = 12x + 350, where x is the number of dogs groomed in a month. Find (R − C)(x). What does it represent? Find (R − C)(20).

Part 3 · Partner practice

Partner 1

Let f(x) = $4x^2 - x$ and g(x) = $x^2 + 3x - 6$. Find (f − g)(x) and (f − g)(2).

Partner 2

Let f(x) = 12x and g(x) = 3x^($\frac{1}{2}$). Find ($\frac{f}{g}$)(x) and its domain.